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## 1. Develop a dashboard using Pandas and Matplotlib to monitor district-wise health indicators.SDG 3

```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files

uploaded = files.upload()

df = pd.read_csv("health_data.csv")

print("----- HEALTH DATASET -----")
display(df)

print("\n----- DATASET INFORMATION -----")
print(df.info())

print("\n----- MISSING VALUES -----")
print(df.isnull().sum())

numeric_columns = df.select_dtypes(include="number").columns

df[numeric_columns] = df[numeric_columns].fillna(
    df[numeric_columns].mean()
)

print("\nMissing values after preprocessing:")
print(df.isnull().sum())

print("\n----- STATISTICAL SUMMARY -----")
display(df.describe())

print("\n----- AVERAGE HEALTH INDICATORS -----")
```

```
print("Average Infant Mortality Rate:",
      round(df["Infant_Mortality_Rate"].mean(), 2))

print("Average Immunization Rate:",
      round(df["Immunization_Rate"].mean(), 2), "%")

print("Average Life Expectancy:",
      round(df["Life_Expectancy"].mean(), 2), "years")

print("Total Hospitals:",
      df["Hospitals"].sum())

print("Average Doctors per 1000:",
      round(df["Doctors_per_1000"].mean(), 2))

best_immunization = df.loc[
    df["Immunization_Rate"].idxmax(), "District"
]

worst_immunization = df.loc[
    df["Immunization_Rate"].idxmin(), "District"
]

lowest_infant_mortality = df.loc[
    df["Infant_Mortality_Rate"].idxmin(), "District"
]

highest_infant_mortality = df.loc[
    df["Infant_Mortality_Rate"].idxmax(), "District"
]

print("\n----- DISTRICT ANALYSIS -----")
```

```
worst_immunization
lowest_infant_mortality

print("Lowest Infant Mortality Rate:",
      lowest_infant_mortality)

print("Highest Infant Mortality Rate:",
      highest_infant_mortality)

fig = plt.figure(figsize=(18, 12))

fig.suptitle(
    "District-Wise Health Indicators Dashboard - SDG 3",
    fontsize=20,
    fontweight="bold"
)

ax1 = plt.subplot(2, 3, 1)

ax1.bar(
    df["District"],
    df["Immunization_Rate"]
)

ax1.set_title("District-wise Immunization Rate")
ax1.set_xlabel("District")
ax1.set_ylabel("Immunization Rate (%)")

ax1.tick_params(axis="x", rotation=45)

ax2 = plt.subplot(2, 3, 2)

ax2.bar(
    df["District"],
    df["Infant_Mortality_Rate"]
)
```

```
print("\n===== KEY FINDINGS =====")
print("\n=====")

print(
    f"1. {best_immunization} has the highest "
    f"immunization rate of "
    f"{df['Immunization_Rate'].max()}%."
)

print(
    f"2. {worst_immunization} has the lowest "
    f"immunization rate of "
    f"{df['Immunization_Rate'].min()}%."
)

print(
    f"3. {lowest_infant_mortality} has the lowest "
    f"infant mortality rate of "
    f"{df['Infant_Mortality_Rate'].min()}%."
)

print(
    f"4. {highest_infant_mortality} has the highest "
    f"infant mortality rate of "
    f"{df['Infant_Mortality_Rate'].max()}%."
)

print(
    f"5. The district with the highest number of hospitals "
    f"is {df.loc[df['Hospitals'].idxmax(), 'District']}."
)

print()
```

```
*** health_data.csv
health_data.csv (text/csv) - 423 bytes, last modified: 8/31/2026 - 100% done
Saving health_data.csv to health_data (1).csv
===== HEALTH DATASET =====
```

|   | District        | Maternal_Mortality_Rate | Infant_Mortality_Rate | Immunization_Rate | Life_Expectancy | Hospitals | Doctors_per_1000 |
|---|-----------------|-------------------------|-----------------------|-------------------|-----------------|-----------|------------------|
| 0 | Chennai         | 52                      | 12                    | 96                | 75.2            | 48        | 2.8              |
| 1 | Coimbatore      | 61                      | 15                    | 93                | 74.6            | 35        | 2.3              |
| 2 | Madurai         | 68                      | 18                    | 91                | 73.8            | 28        | 1.9              |
| 3 | Salem           | 73                      | 21                    | 89                | 72.9            | 25        | 1.7              |
| 4 | Tiruchirappalli | 59                      | 16                    | 94                | 74.1            | 31        | 2.1              |
| 5 | Tirunelveli     | 76                      | 23                    | 87                | 72.5            | 22        | 1.5              |
| 6 | Erode           | 55                      | 14                    | 95                | 75.0            | 27        | 2.2              |
| 7 | Vellore         | 70                      | 20                    | 90                | 73.2            | 30        | 1.8              |
| 8 | Thanjavur       | 64                      | 17                    | 92                | 74.0            | 24        | 2.0              |
| 9 | Thoothukudi     | 58                      | 13                    | 94                | 74.8            | 26        | 2.4              |

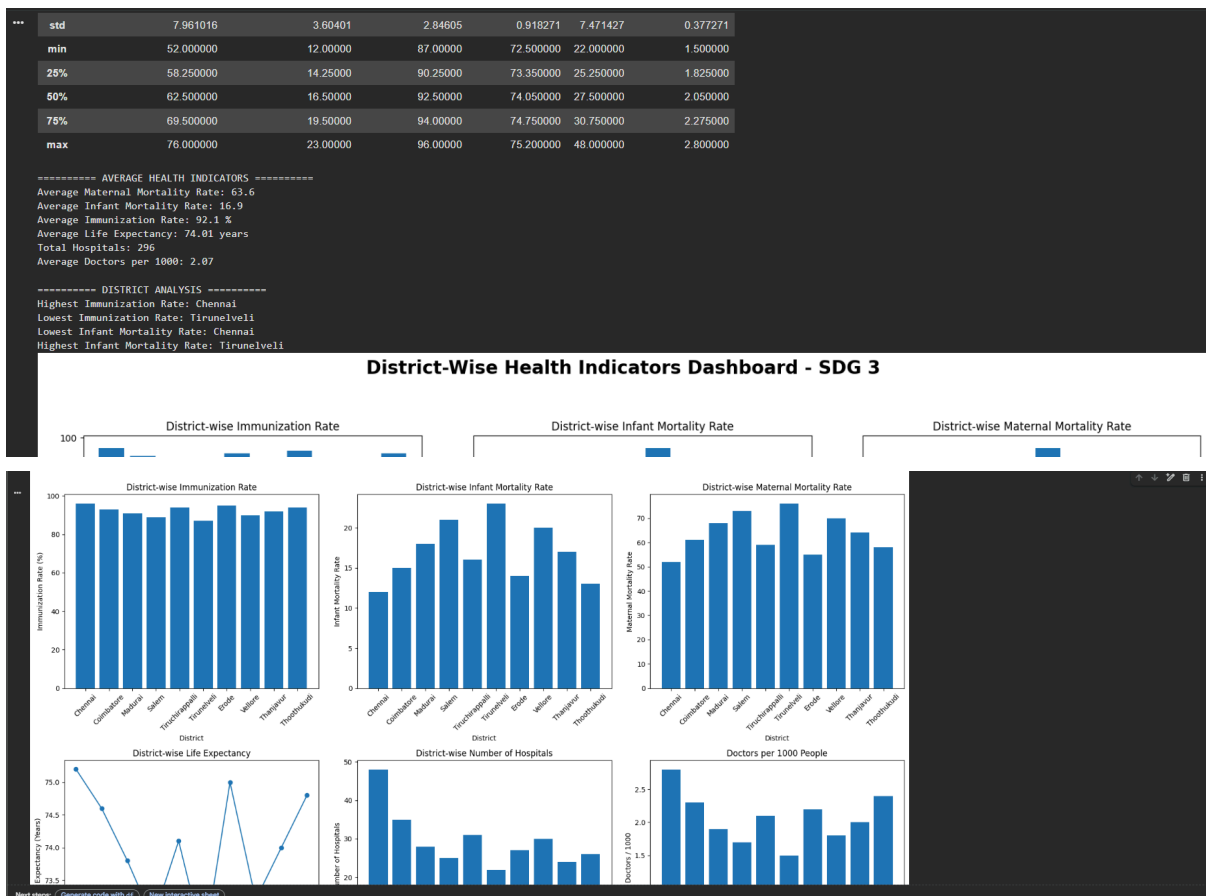
```
===== DATASET INFORMATION =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   District              10 non-null    object
1   Maternal_Mortality_Rate 10 non-null    float64
2   Infant_Mortality_Rate  10 non-null    float64
3   Immunization_Rate      10 non-null    float64
4   Life_Expectancy        10 non-null    float64
5   Hospitals              10 non-null    int64
6   Doctors_per_1000       10 non-null    float64
dtypes: float64(6), int64(1), object(1)
memory usage: 692.0+ bytes
None
```

```
===== MISSING VALUES =====
District              0
Maternal_Mortality_Rate 0
Infant_Mortality_Rate  0
Immunization_Rate     0
Life_Expectancy       0
Hospitals             0
Doctors_per_1000      0
dtype: int64

Missing values after preprocessing:
District              0
Maternal_Mortality_Rate 0
Infant_Mortality_Rate  0
Immunization_Rate     0
Life_Expectancy       0
Hospitals             0
Doctors_per_1000      0
dtype: int64

===== STATISTICAL SUMMARY =====
```

|       | Maternal_Mortality_Rate | Infant_Mortality_Rate | Immunization_Rate | Life_Expectancy | Hospitals | Doctors_per_1000 |
|-------|-------------------------|-----------------------|-------------------|-----------------|-----------|------------------|
| count | 10.000000               | 10.000000             | 10.000000         | 10.000000       | 10.000000 | 10.000000        |
| mean  | 63.600000               | 16.900000             | 92.100000         | 74.010000       | 29.600000 | 2.070000         |
| std   | 7.961016                | 3.604011              | 2.846055          | 0.918271        | 7.471427  | 0.377271         |



## 2. Develop a comparative analysis of renewable energy adoption across Indian states using visualization techniques. SDG 7 & 13

## 2. Comparative Analysis of Renewable Energy Adoption Across Indian States – SDG 7 & 13

### Objective:

The aim is to compare renewable energy adoption across different Indian states using Pandas and Matplotlib. The analysis focuses on installed renewable energy capacity and the contribution of renewable sources, supporting SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action).

For an assignment/demo, we can use a sample dataset containing states and renewable-energy indicators.

### Key indicators

- State
- Solar Capacity (MW)
- Wind Capacity (MW)

- Hydropower Capacity (MW)
- Biomass Capacity (MW)
- Total Renewable Capacity (MW)

## Comparative Analysis

The dataset can be analyzed to identify:

1. Which state has the highest total renewable-energy capacity.
2. Which states have greater dependence on solar or wind energy.
3. Differences in renewable-energy adoption between states.
4. The contribution of different renewable sources.
5. States that could increase renewable adoption to support climate-action goals.

## Visualization Techniques

Using Matplotlib, we can create:

- Bar chart – Compare total renewable capacity between states.
- Stacked bar chart – Compare solar, wind, hydro and biomass contributions.
- Pie chart – Show the overall contribution of different renewable sources.
- Scatter plot – Study the relationship between solar and wind capacity.

## Example Visualization

### Conclusion:

This comparative analysis helps identify differences in renewable-energy adoption across Indian states. States with higher renewable capacity demonstrate greater progress toward SDG 7, while increased renewable-energy use can reduce dependence on fossil fuels and greenhouse-gas emissions, supporting SDG 13. Pandas enables efficient data processing and comparison, while Matplotlib provides clear visualizations for interpreting the results.

## Renewable Energy Capacity by Indian State

Illustrative comparison of total installed renewable-energy capacity across selected Indian states.

| state      | total  |
|------------|--------|
| Rajasthan  | 25,000 |
| Tamil Nadu | 18,000 |
| Karnataka  | 16,500 |
| Gujarat    | 15,500 |

|                |        |
|----------------|--------|
| Andhra Pradesh | 11,000 |
| Maharashtra    | 10,500 |
| Madhya Pradesh | 8,500  |
| Telangana      | 6,000  |